Bandwidth Requirements

Bandwidth Requirements Reference

Proper bandwidth planning is essential for reliable video surveillance. This reference provides the formulas and metrics needed to calculate capacity for your GCXONE deployment.

Calculation Formula

Use this formula to estimate the total bandwidth required for your installation:

Total Bandwidth =
(Number of Cameras × Avg Bitrate) +
(Concurrent Viewers × Streaming Bitrate) +
30% Management Overhead

Bitrate Guidelines

Bandwidth consumption varies significantly based on resolution and the codec used. **H.265 (HEVC)** is highly recommended for 50% bandwidth savings.

| Resolution | Frame Rate | Typical Bitrate | | :--- | :--- | :--- | | **720p (HD)** | 30 fps | 1.5 Mbps | | **1080p (2MP)** | 30 fps | 3.0 Mbps | | **1440p (4MP)** | 30 fps | 5.0 Mbps | | **4K (8MP)** | 30 fps | 12.0 Mbps | | Resolution | Frame Rate | Typical Bitrate | | :--- | :--- | :--- | | **720p (HD)** | 30 fps | 3.0 Mbps | | **1080p (2MP)** | 30 fps | 6.0 Mbps | | **1440p (4MP)** | 30 fps | 10.0 Mbps | | **4K (8MP)** | 30 fps | 25.0 Mbps |

Installation Examples

🏠 Small (10 Cameras)

- **Resolution:** 1080p (H.265)
- **Upload Required:** ~30 Mbps
- **Viewers:** 2 Concurrent Streams
- **Total Recommended: 50 Mbps Symmetric**

🏢 Medium (50 Cameras)

- **Resolution:** Mixed 1080p/4MP (H.265)

- **Upload Required:** ~200 Mbps
- **Viewers:** 10 Concurrent Streams
- **Total Recommended: 400 Mbps Symmetric**

🏭 Large (200+ Cameras)

- **Resolution:** Mixed (H.265)
- **Upload Required:** ~1 Gbps
- **Total Recommended: 2 Gbps Dedicated Link**

Optimization Strategies

- **Use H.265 Everywhere:** Ensure your cameras and NVRs are set to HEVC/H.265 to cut data costs.
- **Motion-Based Bitrate:** Configure **Variable Bitrate (VBR)** so the stream only spikes during actual activity.
- **Sub-Streaming:** Use a lower resolution "Sub-Stream" for multi-camera grid views, and only switch to "Main Stream" for full-screen inspection.
- **VLAN Segmentation:** Keep video traffic on a dedicated VLAN to prevent congestion with general office data.

Common Issues

- **Freezing/Artifacts:** Usually caused by sustained upload congestion. Try reducing the frame rate from 30 fps to 15 fps.
- **Delayed Alarms:** Often a result of network saturation. Implement **Quality of Service (QoS)** to prioritize alarm handshake traffic (Port 443).

Related Articles

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